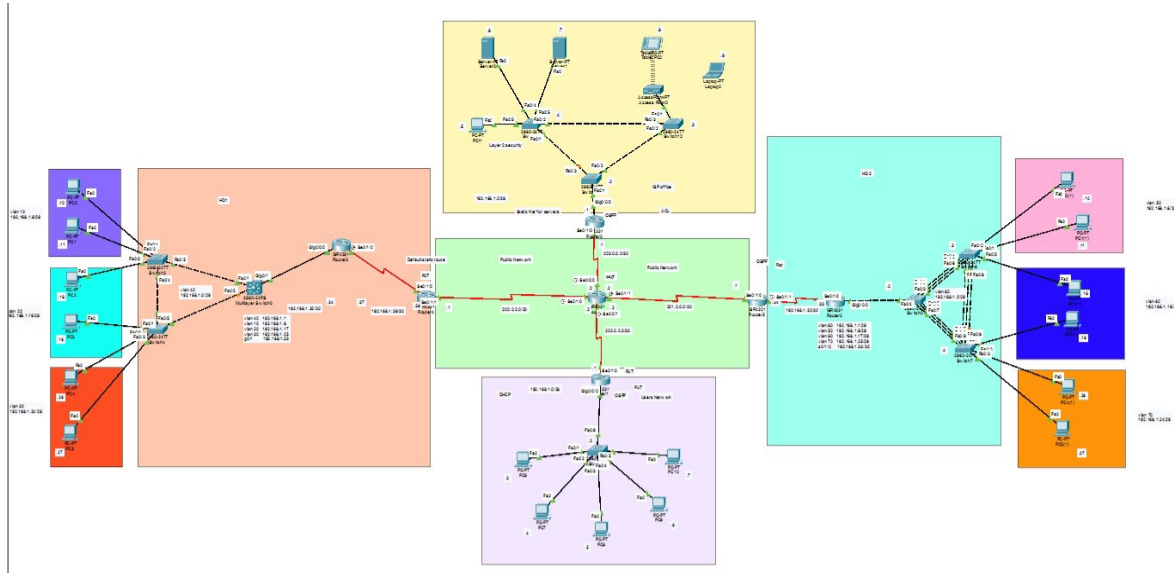


# SUPPLY CHAIN NETWORK

## ARCHITECTURE & IMPLEMENTATION DOCUMENTATION



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# 1. Project Overview

## 1.a Project Description (Supply Chain Network)

This document presents the complete design, addressing plan, device configuration, and verification results for a multi-site **Supply Chain Network** built and simulated in Cisco Packet Tracer. The network models a company with two headquarters sites (HQ1 and HQ2), a dedicated Users/DHCP network, and an ISP Office / Server Farm site, all interconnected through a central core router over a simulated public WAN using OSPF as the dynamic routing protocol.

## 1.b Network Objectives

- Provide logical segmentation of each branch's LAN using VLANs to separate departments/traffic types.
- Provide Layer 3 inter-VLAN routing at each site using either a Multilayer Switch (HQ1) or Router-on-a-Stick (HQ2).
- Interconnect all sites over a simulated public WAN using OSPF as a single-area (Area 0) dynamic routing protocol.
- Translate private addressing to public addressing at each site's edge using NAT/PAT (overload) and Static NAT for published servers.
- Provide automatic IP assignment for the Users Network via a DHCP server configured on its gateway router.
- Secure administrative access to every device using encrypted passwords (enable secret) and SSH for remote management.
- Improve link redundancy and available bandwidth inside HQ2 using Layer 2 EtherChannel (Port-Channel) bundles.
- Verify end-to-end reachability, NAT translation, OSPF neighbor adjacency, and DHCP operation through hands-on testing.

## 1.c Network Topology & Zones Overview

The network is organized into five functional zones, all meeting at a central core router (**Router6**) which is the hub of the simulated public/OSPF WAN:

| Zone                  | Core Devices                                                            | Function                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HQ1                   | Switch3, Switch4, Multilayer Switch0 (MLS), Router3, Router4            | Main HQ LAN with VLANs 10/20/30, inter-VLAN routing on the Multilayer Switch, PAT to WAN via Router4                                                                 |
| HQ2                   | Switch0, Switch6, Switch7, Router2, Router5                             | Secondary HQ LAN with VLANs 50/60/70, Router-on-a-Stick inter-VLAN routing on Router2, EtherChannel between access and distribution switches, PAT to WAN via Router5 |
| Users Network         | Router1, single flat switch, PC6-PC10                                   | Flat /28 network with DHCP-assigned addressing, NAT/PAT to WAN via Router1                                                                                           |
| ISP Office / Servers  | Switch10, Switch11, Switch12, Router0, Server0, Server1, Tablet, Laptop | Publishes two internal servers to the public network using Static NAT; hosts wireless/mobile clients                                                                 |
| Core / Public Network | Router6                                                                 | Central hub router; terminates 4 point-to-point WAN links (200-203.0.0/30) to the four edge routers and runs OSPF Area 0 across all of them                          |

Each of the four branch/site routers (Router0, Router1, Router4, Router5) connects to Router6 over a dedicated point-to-point Serial WAN link using a /30 subnet from the 200-203.0.0.0 range, and performs **NAT overload (PAT)** or **Static NAT** on that link so that internal 192.168.1.0/24 addressing (reused independently at each site)

is translated before reaching the shared core/public network.

## 2. Network Design and Addressing

### 2.a Network Segmentation & VLAN Mapping

Both HQ sites use an identical VLAN numbering pattern (offset by site) so that VLAN role is easy to recognize: user VLANs occupy the first three IDs of each site, and the last VLAN ID at each site is reserved as a management/native VLAN.

| VLAN | Site      | Subnet          | Purpose / Hosts                 | Access Switch(es)        |
|------|-----------|-----------------|---------------------------------|--------------------------|
| 10   | HQ1       | 192.168.1.8/29  | PC0, PC1                        | SW3 (Fa0/1, Fa0/2)       |
| 20   | HQ1       | 192.168.1.16/29 | PC2, PC3                        | SW3 (Fa0/3), SW4 (Fa0/1) |
| 30   | HQ1       | 192.168.1.24/29 | PC4, PC5                        | SW4 (Fa0/2, Fa0/3)       |
| 40   | HQ1       | 192.168.1.0/29  | Management (MLS, SW3, SW4 SVIs) | MLS / SW3 / SW4          |
| 50   | HQ2       | 192.168.1.8/29  | PC0, PC1                        | SW6 (Fa0/1, Fa0/2)       |
| 60   | HQ2       | 192.168.1.16/29 | PC2, PC3                        | SW6 (Fa0/3), SW7 (Fa0/1) |
| 70   | HQ2       | 192.168.1.24/29 | PC4, PC5                        | SW7 (Fa0/2, Fa0/3)       |
| 80   | HQ2       | 192.168.1.0/29  | Management (SW0, SW6, SW7 SVIs) | SW0 / SW6 / SW7          |
| 99   | HQ1 & HQ2 | -               | Native / trunk-only VLAN        | All trunk links          |

■■■■■■■: VLAN 800 appears in the allowed-VLAN list of one trunk on Switch6 (Fa0/4-0/6) but is not provisioned anywhere else in the captured configuration (no VLAN 800 SVI, no access ports). It is flagged here as a configuration item to review/clean up - see Section 7.c Recommendations.

### 2.b Subnetting for Networks

All LAN subnets are carved out of the private block **192.168.1.0/24**, re-used independently at each site behind NAT. Point-to-point WAN/backbone links use /30 subnets. The full subnetting worksheet below lists every block in the design with its mask, wildcard mask, usable host range, and broadcast address.

| Network ID / CIDR | Subnet Mask     | Wildcard Mask | Usable Host Range  | Broadcast    | Used For                                  |
|-------------------|-----------------|---------------|--------------------|--------------|-------------------------------------------|
| 192.168.1.0/28    | 255.255.255.240 | 0.0.0.15      | 192.168.1.1 - .14  | 192.168.1.15 | Users Network (Router1 LAN, DHCP)         |
| 192.168.1.0/29    | 255.255.255.248 | 0.0.0.7       | 192.168.1.1 - .6   | 192.168.1.7  | VLAN 40 (HQ1) / VLAN 80 (HQ2) - Mgmt      |
| 192.168.1.8/29    | 255.255.255.248 | 0.0.0.7       | 192.168.1.9 - .14  | 192.168.1.15 | VLAN 10 (HQ1) / VLAN 50 (HQ2)             |
| 192.168.1.16/29   | 255.255.255.248 | 0.0.0.7       | 192.168.1.17 - .22 | 192.168.1.23 | VLAN 20 (HQ1) / VLAN 60 (HQ2)             |
| 192.168.1.24/29   | 255.255.255.248 | 0.0.0.7       | 192.168.1.25 - .30 | 192.168.1.31 | VLAN 30 (HQ1) / VLAN 70 (HQ2)             |
| 192.168.1.32/30   | 255.255.255.252 | 0.0.0.3       | 192.168.1.33 - .34 | 192.168.1.35 | MLS-Router3 (HQ1) / Router2-Router5 (HQ2) |
| 192.168.1.36/30   | 255.255.255.252 | 0.0.0.3       | 192.168.1.37 - .38 | 192.168.1.39 | Router3 - Router4 (HQ1 internal link)     |

| Network ID / CIDR | Subnet Mask     | Wildcard Mask | Usable Host Range | Broadcast  | Used For                              |
|-------------------|-----------------|---------------|-------------------|------------|---------------------------------------|
| 200.0.0.0/30      | 255.255.255.252 | 0.0.0.3       | 200.0.0.1 - .2    | 200.0.0.3  | Router4 (PAT/HQ1) - Router6 (Core)    |
| 201.0.0.0/30      | 255.255.255.252 | 0.0.0.3       | 201.0.0.1 - .2    | 201.0.0.3  | Router5 (PAT/HQ2) - Router6 (Core)    |
| 202.0.0.0/30      | 255.255.255.252 | 0.0.0.3       | 202.0.0.1 - .2    | 202.0.0.3  | Router1 (Users Net) - Router6 (Core)  |
| 203.0.0.0/30      | 255.255.255.252 | 0.0.0.3       | 203.0.0.1 - .2    | 203.0.0.3  | Router0 (ISP Office) - Router6 (Core) |
| 204.0.0.0/28      | 255.255.255.240 | 0.0.0.15      | 204.0.0.1 - .14   | 204.0.0.15 | Static NAT pool (Server0 / Server1)   |

## 2.c Router & Multilayer Switch IP Addressing Table

| Device          | Interface                    | IP Address                            | Notes                                        |
|-----------------|------------------------------|---------------------------------------|----------------------------------------------|
| Router0         | GigabitEthernet0/0/0         | 192.168.1.1/28                        | ISP Office LAN (NAT inside)                  |
| Router0         | Serial0/1/0                  | 203.0.0.1/30                          | To Router6 (NAT outside)                     |
| Router1         | GigabitEthernet0/0/0         | 192.168.1.1/28                        | Users Network LAN (NAT inside, DHCP gateway) |
| Router1         | Serial0/1/0                  | 202.0.0.1/30                          | To Router6 (NAT outside)                     |
| Router2         | Gi0/0/0.50 / .60 / .70 / .80 | 192.168.1.9/29, .17/29, .25/29, .1/29 | HQ2 VLAN gateways (sub-interfaces)           |
| Router2         | Serial0/1/0                  | 192.168.1.33/30                       | To Router5                                   |
| Router3         | GigabitEthernet0/0/0         | 192.168.1.34/30                       | To Multilayer Switch0 (Gi0/1)                |
| Router3         | Serial0/1/0                  | 192.168.1.37/30                       | To Router4 (DCE, clock rate 2000000)         |
| Router4         | Serial0/1/0                  | 192.168.1.38/30                       | To Router3 (NAT inside)                      |
| Router4         | Serial0/1/1                  | 200.0.0.1/30                          | To Router6 (NAT outside)                     |
| Router5         | Serial0/1/1                  | 192.168.1.34/30                       | To Router2 (NAT inside, DCE)                 |
| Router5         | Serial0/1/0                  | 201.0.0.1/30                          | To Router6 (NAT outside)                     |
| Router6 (Core)  | Serial0/1/0                  | 200.0.0.2/30                          | To Router4 (DCE)                             |
| Router6 (Core)  | Serial0/1/1                  | 201.0.0.2/30                          | To Router5 (DCE)                             |
| Router6 (Core)  | Serial0/2/0                  | 203.0.0.2/30                          | To Router0 (DCE)                             |
| Router6 (Core)  | Serial0/2/1                  | 202.0.0.2/30                          | To Router1 (DCE)                             |
| MLS (3560-24PS) | VLAN10 / 20 / 30 / 40 (SVIs) | 192.168.1.9/29, .17/29, .25/29, .1/29 | HQ1 VLAN gateways                            |
| MLS (3560-24PS) | GigabitEthernet0/1           | 192.168.1.33/30                       | To Router3 (routed port, no switchport)      |

## 3. Layer 2 Switch Configuration

### 3.a Switch Management IP and Default Gateway

Access-layer switches at HQ1 and HQ2 are managed via a VLAN interface (SVI) on the site's dedicated management VLAN (VLAN 40 at HQ1, VLAN 80 at HQ2). Switches at the ISP Office site use the default VLAN 1 for management, addressed out of the site's /28 block.

| Switch | Site       | Mgmt VLAN | Management IP                    |
|--------|------------|-----------|----------------------------------|
| SW3    | HQ1        | VLAN 40   | 192.168.1.2/29                   |
| SW4    | HQ1        | VLAN 40   | 192.168.1.3/29                   |
| MLS    | HQ1        | VLAN 40   | 192.168.1.1/29 (also L3 gateway) |
| SW0    | HQ2        | VLAN 80   | 192.168.1.2/29                   |
| SW6    | HQ2        | VLAN 80   | 192.168.1.3/29                   |
| SW7    | HQ2        | VLAN 80   | 192.168.1.4/29                   |
| SW10   | ISP Office | VLAN 1    | 192.168.1.2/28                   |
| SW11   | ISP Office | VLAN 1    | 192.168.1.4/28                   |
| SW12   | ISP Office | VLAN 1    | 192.168.1.3/28                   |

■■■■■■■: Default gateway for each management SVI is the site's local Layer-3 device (MLS at HQ1, Router2 via Gi0/0/0.80 at HQ2). SW10/11/12 remain on VLAN1 with no explicit VLAN segmentation configured on their access ports - see Recommendations.

### 3.b VLAN Database & Port Allocations (Access/Trunk)

| Switch    | Port(s)      | Configuration                                            |
|-----------|--------------|----------------------------------------------------------|
| SW3 (HQ1) | Fa0/1, Fa0/2 | Access - VLAN 10                                         |
| SW3 (HQ1) | Fa0/3        | Access - VLAN 20                                         |
| SW3 (HQ1) | Fa0/4, Fa0/5 | Trunk - native 99, allowed 10,20,30,40,99                |
| SW4 (HQ1) | Fa0/1        | Access - VLAN 20                                         |
| SW4 (HQ1) | Fa0/2, Fa0/3 | Access - VLAN 30                                         |
| SW4 (HQ1) | Fa0/4, Fa0/5 | Trunk - native 99, allowed 10,20,30,40,99                |
| MLS (HQ1) | Fa0/1        | Trunk - native 99, allowed 10,20,30,40,99                |
| SW6 (HQ2) | Fa0/1, Fa0/2 | Access - VLAN 50                                         |
| SW6 (HQ2) | Fa0/3        | Access - VLAN 60                                         |
| SW6 (HQ2) | Fa0/4-Fa0/6  | Trunk (Po1) - native 99, allowed 50,60,70,99,800         |
| SW6 (HQ2) | Fa0/7-Fa0/9  | Trunk - native 99, allowed 50,60,70,80,99 (to SW7)       |
| SW7 (HQ2) | Fa0/1        | Access - VLAN 60                                         |
| SW7 (HQ2) | Fa0/2, Fa0/3 | Access - VLAN 70                                         |
| SW7 (HQ2) | Fa0/4-Fa0/6  | Trunk (Po3) - native 99, allowed 50,60,70,80,99          |
| SW7 (HQ2) | Fa0/7-Fa0/9  | Trunk (Po2) - native 99, allowed 50,60,70,80,99 (to SW0) |

| Switch    | Port(s)     | Configuration                                            |
|-----------|-------------|----------------------------------------------------------|
| SW0 (HQ2) | Fa0/1-Fa0/3 | Trunk (Po1) - native 99, allowed 50,60,70,80,99 (to SW6) |
| SW0 (HQ2) | Fa0/5-Fa0/7 | Trunk (Po2) - native 99, allowed 50,60,70,80,99 (to SW7) |

### 3.c EtherChannel Configuration

HQ2 uses Layer 2 EtherChannel (PAgP/LACP-style **active** mode) to bundle parallel links between the distribution switch (SW0) and each access switch, increasing available bandwidth and providing link redundancy:

| Bundle         | Switch A Side   | Switch B Side     | Mode                        |
|----------------|-----------------|-------------------|-----------------------------|
| Port-channel 1 | SW0 (Fa0/1-0/3) | SW6 (Fa0/4-0/6)   | channel-group 1 mode active |
| Port-channel 2 | SW0 (Fa0/5-0/7) | SW7 (Fa0/7-0/9)   | channel-group 2 mode active |
| Port-channel 3 | SW7 (Fa0/4-0/6) | SW6 (Fa0/7-0/9) * | channel-group 3 mode active |

■■■■■■: (\*) SW7's Fa0/4-0/6 are configured with **channel-group 3**, but the corresponding ports on SW6 (Fa0/7-0/9) were captured as a plain trunk with **no channel-group command**. This is a one-sided EtherChannel configuration and should be corrected on SW6 for Port-channel 3 to negotiate correctly - see Recommendations.

### 3.d Core Switch Configuration

HQ1's core/distribution device is a **Layer 3 Multilayer Switch (3560-24PS, hostname MLS)** that performs both VLAN switching and inter-VLAN routing:

- **ip routing** is enabled globally to allow Layer-3 forwarding between VLANs.
- Fa0/1 is configured as an 802.1Q trunk (native VLAN 99, allowed VLANs 10,20,30,40,99) toward the HQ1 access switches.
- Gi0/1 is converted to a routed port (**no switchport**) addressed 192.168.1.33/30 toward Router3.
- Four SVIs (VLAN10, 20, 30, 40) act as the default gateway for each HQ1 VLAN.
- OSPF Area 0 is enabled and advertises all five directly connected subnets, allowing the MLS to exchange routes with Router3 and, transitively, the rest of the network.

## 4. Routing Protocols & NAT

### 4.a Router Interface & Inter-VLAN Configuration (Sub-interfaces)

HQ2 uses the classic **Router-on-a-Stick** model: a single physical link (Gi0/0/0) from Router2 to the HQ2 switch stack carries all VLAN traffic as an 802.1Q trunk, and Router2 performs inter-VLAN routing using one sub-interface per VLAN:

```
interface GigabitEthernet0/0/0
  no ip address
!
interface GigabitEthernet0/0/0.10
  encapsulation dot1Q 10
  no ip address                                ! defined, not yet in service
!
interface GigabitEthernet0/0/0.50
  encapsulation dot1Q 50
  ip address 192.168.1.9 255.255.255.248
!
interface GigabitEthernet0/0/0.60
  encapsulation dot1Q 60
  ip address 192.168.1.17 255.255.255.248
!
interface GigabitEthernet0/0/0.70
  encapsulation dot1Q 70
  ip address 192.168.1.25 255.255.255.248
!
interface GigabitEthernet0/0/0.80
  encapsulation dot1Q 80
  ip address 192.168.1.1 255.255.255.248
```

■■■■■■■: A live CLI capture shows an attempt to bring up sub-interface .10 with 192.168.1.1/29, which failed with "192.168.1.0 overlaps with GigabitEthernet0/0/0.10" because that block is already owned by VLAN 80. Sub-interface .10 (VLAN 10) is therefore defined but has **no IP address configured** and is not currently passing routed traffic - it exists in the trunk's allowed list only. HQ1 achieves the equivalent function using a Multilayer Switch instead of Router-on-a-Stick (Section 3.d).

### 4.b Default Static Routing Configuration

Each branch edge router that performs PAT to the core uses a single default static route pointing out its WAN-facing serial interface, since the core network is reachable only through that one path:

| Router  | Static Route                             | Purpose                                         |
|---------|------------------------------------------|-------------------------------------------------|
| Router1 | ip route 0.0.0.0 0.0.0.0 Serial0/1/0     | toward Router6                                  |
| Router4 | ip route 0.0.0.0 0.0.0.0 Serial0/1/1     | toward Router6                                  |
| Router5 | ip route 0.0.0.0 0.0.0.0 Serial0/1/0     | toward Router6                                  |
| Router0 | ip route 204.0.0.0 255.255.255.240 Null0 | discard route for the published Static-NAT pool |

Each of these routers also uses **default-information originate** under OSPF so this default route is redistributed into the OSPF domain, giving every internal router a path to the Internet-facing edge.

### 4.c Dynamic Routing (OSPF Configuration)

A single OSPF process (process ID 1, **Area 0**) runs across the entire network - on every router and on the HQ1 Multilayer Switch - providing full reachability between all sites without manual route maintenance.

| Device  | Networks Advertised (area 0) | Key Options                 |
|---------|------------------------------|-----------------------------|
| Router0 | 192.168.1.0/28, 203.0.0.0/30 | redistribute static subnets |

| Device         | Networks Advertised (area 0)                           | Key Options                                                 |
|----------------|--------------------------------------------------------|-------------------------------------------------------------|
| Router1        | 192.168.1.0/28                                         | passive-interface Se0/1/0,<br>default-information originate |
| Router2        | 192.168.1.0/29, .8/29, .16/29, .24/29, 192.168.1.32/30 | -                                                           |
| Router3        | 192.168.1.32/30, 192.168.1.36/30                       | -                                                           |
| Router4        | 192.168.1.36/30                                        | passive-interface Se0/1/1,<br>default-information originate |
| Router5        | 192.168.1.32/30                                        | passive-interface Se0/1/0,<br>default-information originate |
| Router6 (Core) | 200.0.0.0/30, 201.0.0.0/30, 202.0.0.0/30, 203.0.0.0/30 | backbone hub - 4 neighbors                                  |
| MLS            | 192.168.1.0/29, .8/29, .16/29, .24/29, 192.168.1.32/30 | -                                                           |

**passive-interface** is applied on the WAN-facing (NAT-outside) interface of each edge router so that OSPF Hellos are not sent toward the core over the same link that Router6 also terminates - preventing unnecessary adjacency flapping while still advertising the subnet.

## 4.d NAT / PAT (Overload) Configuration

Three routers (Router1, Router4, Router5) perform dynamic **PAT (NAT Overload)**, translating their entire internal 192.168.1.0/24-based subnet to a single public IP on their WAN interface:

```
access-list 1 permit 192.168.1.0 0.0.0.255
!
ip nat inside source list 1 interface Serial0/1/x overload
!
interface <LAN-facing interface>
 ip nat inside
!
interface <WAN-facing interface>
 ip nat outside
```

Router4 additionally shows an **ip nat pool PAT\_POOL** (200.0.0.1 200.0.0.1 /30) defined during configuration, though the final active NAT statement uses the simpler **interface ... overload** form rather than the pool.

Router0 (ISP Office) instead uses **Static NAT** to permanently publish two internal servers to the public network:

```
ip nat inside source static 192.168.1.6 204.0.0.1    ! Server0
ip nat inside source static 192.168.1.7 204.0.0.2    ! Server1
```

## 4.e DHCP Server & Pool Configurations

Router1 provides automatic addressing for the Users Network via a locally-defined DHCP pool:

```
ip dhcp pool users-pool
 network 192.168.1.0 255.255.255.240
 default-router 192.168.1.1
 dns-server 9.9.9.9
```

PCs 6 through 10 in the Users Network obtain their IP address, default gateway (192.168.1.1), and DNS server (9.9.9.9) automatically from this pool.

## 5. Network Management & Security

### 5.a Administrative Access Security (Console & Enable Secret)

- All routers and switches use an MD5-hashed **enable secret** (type 5) rather than a plaintext enable password, except Router5 and Switch7 which were found using **enable password cisco** (plaintext) - flagged in Recommendations.
- A local user account (**username cisco secret <hash>**) is configured on every device for use with **login local** on VTY lines.
- Console line (line con 0) is protected with **password cisco** and **login** on every device.
- VTY line ranges are split across devices (e.g. 0-4 / 5-6 / 5-9), with the tail range on several switches (SW3, SW6, SW7, SW10-12) left on plain **login** instead of **login local** - meaning those extra VTY lines still expect just the line password rather than username+password.
- **ip domain-name cisco.com** is configured on every device (a prerequisite for generating RSA keys / enabling SSH).

### 5.b SSH Remote Access Configuration (Routers & Switches)

SSH was enabled and verified on the core router (Router6); the standard sequence used is:

```
ip domain-name cisco.com
crypto key generate rsa          ! 1024-bit keys generated
username cisco secret cisco
line vty 0 8
  login local
  transport input all
%SSH-5-ENABLED: SSH 1.99 has been enabled
```

A successful remote-login test from a client PC confirmed SSH connectivity into Router1:

```
C:\>ssh -l cisco 192.168.1.1
Password:
R1>
```

Because every device shares the same **ip domain-name**, RSA key generation, and **username cisco secret** pattern seen in the captured running-configs, SSH is expected to be available network-wide, with the console-cable/Telnet fallback (line con 0 / plaintext line password) retained as a secondary access method.

## 6. Testing, Verification & Diagnostics

### 6.a Intra-VLAN and Inter-VLAN Connectivity Testing

End-device connectivity within the same VLAN (e.g. PC0 <-> PC1 on VLAN 10) and across VLANs via each site's Layer-3 gateway (MLS at HQ1, Router2 sub-interfaces at HQ2) was validated using extended pings from the PC command prompt and Packet Tracer's Simulation mode, confirming that inter-VLAN traffic is correctly routed through the local Layer-3 device before reaching another VLAN's subnet.

### 6.b End-to-End Branch-to-Branch Testing (Ping)

ICMP reachability between branch routers over the OSPF-routed core (e.g. PC0 pinging Router6, and cross-branch pings between Router5 and Router6) was exercised through Packet Tracer's PDU list / Simulation panel and returned successful replies, confirming that OSPF Area 0 correctly propagates reachability across all four WAN links.

### 6.c DHCP Verification

Clients in the Users Network (PC6-PC10) were confirmed to receive an address from the **users-pool** DHCP scope (192.168.1.0/28) with default gateway 192.168.1.1 and DNS server 9.9.9.9, as defined on Router1.

### 6.d OSPF Neighbor & Routing Verification

A live capture on Router4 during adjacency formation shows successful OSPF neighbor establishment:

```
04:36:10: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.0.2 on Serial0/1/1 from LOADING to FULL, Loading Done
04:36:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.37 on Serial0/1/0 from LOADING to FULL, Loading Done
This confirms Router4 formed FULL adjacencies both toward the core (Router6, via Serial0/1/1) and toward Router3 (via Serial0/1/0), i.e. OSPF is fully operational end-to-end at this branch.
```

### 6.e NAT Translation Verification

Static NAT for the two published servers was verified from an external client by pinging the public-facing addresses directly, with 100% success:

```
C:\>ping 204.0.0.1
Reply from 204.0.0.1: bytes=32 time=13ms TTL=125
Reply from 204.0.0.1: bytes=32 time=21ms TTL=125
Reply from 204.0.0.1: bytes=32 time=23ms TTL=125
Reply from 204.0.0.1: bytes=32 time=26ms TTL=125
Ping statistics: 4 sent, 4 received, 0% loss
This confirms Router0's static NAT entries (192.168.1.6 ↔ 204.0.0.1 for Server0, and 192.168.1.7 ↔ 204.0.0.2 for Server1) are translating correctly in both directions.

C:\>ping 204.0.0.2
Reply from 204.0.0.2: bytes=32 time=18ms TTL=125
Reply from 204.0.0.2: bytes=32 time=11ms TTL=125
Reply from 204.0.0.2: bytes=32 time=11ms TTL=125
Reply from 204.0.0.2: bytes=32 time=11ms TTL=125
Ping statistics: 4 sent, 4 received, 0% loss
```

### 6.f SSH & Port Security Verification

SSH remote access was verified end-to-end: a PC issuing **ssh -l cisco 192.168.1.1** successfully authenticated and reached the **R1> exec** prompt (see Section 5.b), confirming that RSA key generation, the local user database, and **line vty ... login local / transport input all** are all correctly configured and operational on at least Router6/Router1's management path.

■■■■■■■: No explicit **switchport port-security** commands were captured on any access port during this documentation pass. If port security is a project requirement, it should be added explicitly to each access port (see Recommendations) and re-verified with **show port-security**.

## 7. Conclusion & Appendices

### 7.a Project Summary

This project successfully implements a segmented, multi-site Supply Chain Network covering two headquarters (HQ1, HQ2), a DHCP-served Users Network, and an ISP Office/Server site, all interconnected over a simulated public WAN using single-area OSPF. VLAN segmentation, inter-VLAN routing (via a Multilayer Switch at HQ1 and Router-on-a-Stick at HQ2), NAT/PAT overload for general outbound traffic, Static NAT for two published servers, EtherChannel link bundling at HQ2, and SSH-based administrative access were all configured and verified through direct testing (ping, OSPF adjacency logs, and an interactive SSH session). The network meets its core objectives of segmentation, scalable dynamic routing, controlled address translation, and secured remote management.

### 7.b Device CLI Backup Scripts (Appendix A)

The reconstructed running-configuration for every device documented in this project is provided below.

## Router0 (R0) — ISP Office / Server Farm Gateway - Static NAT

```
hostname R0
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
no ip cef
no ipv6 cef
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
 ip address 192.168.1.1 255.255.255.240
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 no ip address
 shutdown
!
interface GigabitEthernet0/0/2
 no ip address
 shutdown
!
interface Serial0/1/0
 ip address 203.0.0.1 255.255.255.252
 ip nat outside
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 redistribute static subnets
 network 192.168.1.0 0.0.0.15 area 0
 network 203.0.0.0 0.0.0.3 area 0
!
ip nat inside source static 192.168.1.6 204.0.0.1
ip nat inside source static 192.168.1.7 204.0.0.2
ip classless
ip route 204.0.0.0 255.255.255.240 Null0
ip flow-export version 9
!
line con 0
 password cisco
 login
line aux 0
line vty 0 4
 login local
line vty 5
 login local
end
```

## Router1 (R1) — Users Network Gateway - DHCP + PAT

```
hostname R1
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip dhcp pool users-pool
  network 192.168.1.0 255.255.255.240
  default-router 192.168.1.1
  dns-server 9.9.9.9
!
no ip cef
no ipv6 cef
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
  ip address 192.168.1.1 255.255.255.240
  ip nat inside
  duplex auto
  speed auto
!
interface GigabitEthernet0/0/1
  no ip address
  shutdown
!
interface GigabitEthernet0/0/2
  no ip address
  shutdown
!
interface Serial0/1/0
  ip address 202.0.0.1 255.255.255.252
  ip nat outside
!
interface Serial0/1/1
  no ip address
  clock rate 2000000
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  log-adjacency-changes
  passive-interface Serial0/1/0
  network 192.168.1.0 0.0.0.15 area 0
  default-information originate
!
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface Serial0/1/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
ip flow-export version 9
!
line con 0
  password cisco
  login
line aux 0
line vty 0 4
  login local
line vty 5 6
  login local
end
```

## Router2 (R2) — HQ2 LAN Gateway - Router-on-a-Stick

```
hostname R2
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
no ip cef
no ipv6 cef
!
username cisco secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/0/0.10
  encapsulation dot1Q 10
  no ip address
!
interface GigabitEthernet0/0/0.50
  encapsulation dot1Q 50
  ip address 192.168.1.9 255.255.255.248
!
interface GigabitEthernet0/0/0.60
  encapsulation dot1Q 60
  ip address 192.168.1.17 255.255.255.248
!
interface GigabitEthernet0/0/0.70
  encapsulation dot1Q 70
  ip address 192.168.1.25 255.255.255.248
!
interface GigabitEthernet0/0/0.80
  encapsulation dot1Q 80
  ip address 192.168.1.1 255.255.255.248
!
interface GigabitEthernet0/0/1
  no ip address
  shutdown
!
interface GigabitEthernet0/0/2
  no ip address
  shutdown
!
interface Serial0/1/0
  ip address 192.168.1.33 255.255.255.252
!
interface Serial0/1/1
  no ip address
  clock rate 2000000
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  log-adjacency-changes
  network 192.168.1.0 0.0.0.7 area 0
  network 192.168.1.8 0.0.0.7 area 0
  network 192.168.1.16 0.0.0.7 area 0
  network 192.168.1.24 0.0.0.7 area 0
  network 192.168.1.32 0.0.0.3 area 0
!
ip classless
ip flow-export version 9
!
line con 0
```

```
password cisco
login
line aux 0
line vty 0 4
  login local
line vty 5 8
  login local
end
```

### Router3 (R3) — HQ1 LAN Gateway

```
hostname R3
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
no ip cef
no ipv6 cef
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
  ip address 192.168.1.34 255.255.255.252
  duplex auto
  speed auto
!
interface GigabitEthernet0/0/1
  no ip address
  shutdown
!
interface GigabitEthernet0/0/2
  no ip address
  shutdown
!
interface Serial0/1/0
  ip address 192.168.1.37 255.255.255.252
  clock rate 2000000
!
interface Serial0/1/1
  no ip address
  clock rate 2000000
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  log-adjacency-changes
  network 192.168.1.32 0.0.0.3 area 0
  network 192.168.1.36 0.0.0.3 area 0
!
ip classless
ip flow-export version 9
!
line con 0
  password cisco
  login
line aux 0
line vty 0 4
  login local
line vty 5 6
  login local
end
```

## Router4 (R4) — HQ1 Edge Gateway - PAT to Core

```
hostname R4
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
no ip cef
no ipv6 cef
!
username cisco secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
  no ip address
  shutdown
!
interface Serial0/1/0
  ip address 192.168.1.38 255.255.255.252
  ip nat inside
!
interface Serial0/1/1
  ip address 200.0.0.1 255.255.255.252
  ip nat outside
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  log-adjacency-changes
  passive-interface Serial0/1/1
  network 192.168.1.36 0.0.0.3 area 0
  default-information originate
!
ip nat pool PAT_POOL 200.0.0.1 200.0.0.1 netmask 255.255.255.252
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface Serial0/1/1 overload
ip classless
ip route 0.0.0.0 0.0.0.0 Serial0/1/1
ip flow-export version 9
!
line con 0
  password cisco
  login
line aux 0
line vty 0 4
  login local
line vty 5 6
  login local
end
```

## Router5 (R5) — HQ2 Edge Gateway - PAT to Core (enable password NOT encrypted)

```
hostname R5
!
enable password cisco
!
no ip cef
no ipv6 cef
!
username cisco secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
 no ip address
!
interface Serial0/1/0
 ip address 201.0.0.1 255.255.255.252
 ip nat outside
!
interface Serial0/1/1
 ip address 192.168.1.34 255.255.255.252
 ip nat inside
 clock rate 2000000
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 passive-interface Serial0/1/0
 network 192.168.1.32 0.0.0.3 area 0
 default-information originate
!
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface Serial0/1/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
ip flow-export version 9
!
line con 0
 password cisco
 login
line aux 0
line vty 0 4
 login local
line vty 5 9
 login local
end
```

## Router6 (R6) — Core / Public-Network Hub Router - SSH enabled

```
no service password-encryption
hostname R6
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
no ip cef
no ipv6 cef
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
! crypto key generate rsa      (1024-bit, SSH 1.99 enabled)
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
no ip address
shutdown
!
interface Serial0/1/0
ip address 200.0.0.2 255.255.255.252
clock rate 2000000
!
interface Serial0/1/1
ip address 201.0.0.2 255.255.255.252
clock rate 2000000
!
interface Serial0/2/0
ip address 203.0.0.2 255.255.255.252
clock rate 2000000
!
interface Serial0/2/1
ip address 202.0.0.2 255.255.255.252
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 200.0.0.0 0.0.0.3 area 0
network 201.0.0.0 0.0.0.3 area 0
network 202.0.0.0 0.0.0.3 area 0
network 203.0.0.0 0.0.0.3 area 0
!
ip classless
ip flow-export version 9
!
line con 0
password cisco
line aux 0
line vty 0 4
login local
line vty 5 6
login local
transport input all
end
```

## Multilayer Switch0 (MLS) — HQ1 Core L3 Switch - Inter-VLAN Routing

```
hostname MLS
!
no profinet
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip routing
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
spanning-tree mode pvst
!
interface FastEthernet0/1
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  switchport mode trunk
! (FastEthernet0/2 - 0/24 unused / default)
!
interface GigabitEthernet0/1
  no switchport
  ip address 192.168.1.33 255.255.255.252
  duplex auto
  speed auto
! (GigabitEthernet0/2 unused)
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan10
  mac-address 00e0.8fcc.a301
  ip address 192.168.1.9 255.255.255.248
!
interface Vlan20
  mac-address 00e0.8fcc.a302
  ip address 192.168.1.17 255.255.255.248
!
interface Vlan30
  mac-address 00e0.8fcc.a303
  ip address 192.168.1.25 255.255.255.248
!
interface Vlan40
  mac-address 00e0.8fcc.a304
  ip address 192.168.1.1 255.255.255.248
!
router ospf 1
  log-adjacency-changes
  network 192.168.1.0 0.0.0.7 area 0
  network 192.168.1.8 0.0.0.7 area 0
  network 192.168.1.16 0.0.0.7 area 0
  network 192.168.1.24 0.0.0.7 area 0
  network 192.168.1.32 0.0.0.3 area 0
!
ip classless
ip flow-export version 9
!
line con 0
  password cisco
  login
line aux 0
line vty 0 4
  login local
line vty 5
  login local
end
```

## Switch3 (SW3) — HQ1 Access Switch - VLANs 10/20

```
hostname SW3
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/4
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  switchport mode trunk
!
interface FastEthernet0/5
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  switchport mode trunk
! (FastEthernet0/6 - 0/24 unused / default)
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan40
  ip address 192.168.1.2 255.255.255.248
!
line con 0
  password cisco
  login
line vty 0 4
  login local
line vty 5 6
  login local
line vty 7 15
  login
end
```

## Switch4 (SW4) — HQ1 Access Switch - VLANs 20/30

```
hostname SW4
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/4
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  switchport mode trunk
!
interface FastEthernet0/5
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  switchport mode trunk
! (FastEthernet0/6 - 0/24 unused / default)
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan40
  ip address 192.168.1.3 255.255.255.248
!
line con 0
  password cisco
  login
line vty 0 4
  login local
line vty 5 6
  login local
line vty 7 15
  login
end
```

## Switch0 (SW0) — HQ2 Distribution Switch - EtherChannel Po1/Po2

```
hostname SW0
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport trunk allowed vlan 50,60,70,80,99
 switchport mode trunk
!
interface Port-channel2
 switchport trunk native vlan 99
 switchport trunk allowed vlan 50,60,70,80,99
 switchport mode trunk
!
interface FastEthernet0/1
 switchport trunk native vlan 99
 switchport trunk allowed vlan 50,60,70,80,99
 switchport mode trunk
 channel-group 1 mode active
! (Fa0/2, Fa0/3 identical - channel-group 1)
! (Fa0/4 unused)
!
interface FastEthernet0/5
 switchport trunk native vlan 99
 switchport trunk allowed vlan 50,60,70,80,99
 switchport mode trunk
 channel-group 2 mode active
! (Fa0/6, Fa0/7 identical - channel-group 2)
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan80
 ip address 192.168.1.2 255.255.255.248
!
line con 0
 password cisco
 login
line vty 0 4
 login local
line vty 5 8
 login local
line vty 9 15
 login
end
```

## Switch6 (SW6) — HQ2 Access Switch - VLANs 50/60

```
hostname SW6
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/4
  switchport trunk native vlan 99
  switchport trunk allowed vlan 50,60,70,99,800
  switchport mode trunk
  channel-group 1 mode active
! (Fa0/5, Fa0/6 identical - channel-group 1, Po1 toward SW0)
!
interface FastEthernet0/7
  switchport trunk native vlan 99
  switchport trunk allowed vlan 50,60,70,80,99
  switchport mode trunk
! (Fa0/8, Fa0/9 identical trunk - toward SW7, NO channel-group configured here)
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan80
  ip address 192.168.1.3 255.255.255.248
!
line con 0
  password cisco
  login
line vty 0 4
  login local
line vty 5 6
  login local
line vty 7 15
  login
end
```

## Switch7 (SW7) — HQ2 Access Switch - VLANs 60/70 (enable password NOT encrypted)

```
hostname SW7
!
enable password cisco
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel2
  switchport trunk native vlan 99
  switchport trunk allowed vlan 50,60,70,80,99
  switchport mode trunk
!
interface Port-channel3
  switchport trunk mode trunk
!
interface FastEthernet0/1
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 70
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 70
  switchport mode access
!
interface FastEthernet0/4
  switchport trunk native vlan 99
  switchport trunk allowed vlan 50,60,70,80,99
  switchport mode trunk
  channel-group 3 mode active
! (Fa0/5, Fa0/6 identical - channel-group 3, Po3)
!
interface FastEthernet0/7
  switchport trunk native vlan 99
  switchport trunk allowed vlan 50,60,70,80,99
  switchport mode trunk
  channel-group 2 mode active
! (Fa0/8, Fa0/9 identical - channel-group 2, Po2 toward SW0)
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan80
  ip address 192.168.1.4 255.255.255.248
!
line con 0
  password cisco
  login
line vty 0 4
  login local
line vty 5 8
  login local
line vty 9 15
  login
end
```

## Switch10 (SW10) — ISP Office Switch

```
no service password-encryption
hostname SW10
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
! (all FastEthernet0/1-0/24 and Gigabit0/1-0/2 left at default - no VLAN assignment captured)
!
interface Vlan1
 ip address 192.168.1.2 255.255.255.240
!
line con 0
 password cisco
 login
line vty 0 4
 login local
line vty 5 9
 login local
line vty 10 15
 login
end
```

## Switch11 (SW11) — ISP Office Switch

```
no service password-encryption
hostname SW11
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
! (all FastEthernet0/1-0/24 and Gigabit0/1-0/2 left at default - no VLAN assignment captured)
!
interface Vlan1
 ip address 192.168.1.4 255.255.255.240
!
line con 0
 password cisco
 login
line vty 0 4
 login local
line vty 5 8
 login local
line vty 9 15
 login
end
```

## Switch12 (SW12) — ISP Office Switch

```
no service password-encryption
hostname SW12
!
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
ip domain-name cisco.com
!
username cisco secret 5 $1$mERr$hX5rVt7rPNoS4wqbXKX7m0
!
spanning-tree mode pvst
spanning-tree extend system-id
! (all FastEthernet0/1-0/24 and Gigabit0/1-0/2 left at default - no VLAN assignment captured)
!
interface Vlan1
 ip address 192.168.1.3 255.255.255.240
!
line con 0
 password cisco
 login
line vty 0 4
 login local
line vty 5 9
 login local
line vty 10 15
 login
end
```